Specification of Feature Isolation

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1 Basic Information and Results

Feature isolation is based on the following information:

- A specification of a software product line including the names of its independent features, the feature categories to be considered, the maximum rank of interaction between any kind of and- and or-features, and optionally explicitly dependent features plus available systems or missing systems all of which resulting in a set with the identifiers of available systems. This specification is denoted by SPL. An identifier (ID for short) is a natural number with a lower bound of 1.
- For each feature there exists a description that specifies in which systems it occurs and in which it does not plus an associated unique feature ID. Such a feature description is called FD, a feature ID is called FID.
- For each system there exists a description that specifies the features it consists of plus an associated unique system ID. Such a system description is called SD, a system ID is called SID.

The isolation of a feature f_i specified by its FID produces the following outcome:

- A set of FIDs representing the result of the set difference of the set intersection of FIDs of all systems that include f_i (denoted by I) and the set union of FIDs of all systems that do not include f_i (denoted by U). The result set must not be $\emptyset$ (as this would indicate a fundamental error). It can include exactly one element which then must be the FID of f_i. It can include more than one element, but one of them must be the FID of the sought feature f_i. The resulting set difference is denoted by D. A set of minimal set differences of SIDs which evaluate to the result specified before, i.e. D. A set difference is *minimal* if none of its sub expressions includes more elements than required to give D. There must be at least one minimal set difference that gives D. The set of minimal set differences is denoted by MD.

Minimal set difference. The following examples illustrate the meaning of *minimal* in a set difference. Just as a reminder: The elements of these sets are FIDs.

Given are the systems
$SD_1 = \{1, 2, 3\}$,
$SD_2 = \{2, 3, 4\}$,
$SD_3 = \{3, 4, 5\}$,
$SD_4 = \{4, 5, 6\}$,
$SD_5 = \{2, 7\}$ and
$SD_6 = \{1, 8\}$.

With respect to FID_2, the following set intersections I are minimal as they all isolate FID_2.
$I_1 = SD_1 \cap SD_2$

$I_2 = SD_1 \cap SD_5$
$I_3 = SD_2 \cap SD_5$

$I_4 = SD_1 \cap SD_2 \cap SD_5$ is not minimal because I_1, I_2 and I_3 successfully isolate FID_2 but contain less operands. It is important to note that only evaluating I can give the desired result. Furthermore, I cannot be omitted.

With respect to FID_2 the set difference $D_1 = SD_1 \setminus SD_3 \cup SD_6$ is minimal as it isolates FID_2.

$D_2 = SD_1 \cap SD_2 \setminus SD_3$ is not minimal, as I already successfully isolates FID_2. $D_3 = SD_1 \setminus SD_3 \cup SD_4 \cup SD_6$ is also not minimal, as $SD_3 \cup SD_6$ in U already subtracts all unwanted elements of I. It has to be emphasized, that —different from I— U can be omitted.

2 Functions,Datatypes and Variables

The datatypes `feature_ids_t` and `system_ids_t` represent the set types for FIDs and SIDs. The member variable `spl_specification_t::m_available_systems` which is of type `system_ids_t` contains the IDs of the actually available system.